

# Part A: Conceptual Understanding (Theory)

## 1. What are Supervised Learning Algorithms?

Supervised Learning is a type of Machine Learning where the model learns from **labeled data**. We give the model the **input and correct output**, so it learns to predict the output for new data.

**Example:** Predicting marks based on study hours.

## 2. Difference between Regression Algorithms vs Classification Algorithms

**Regression** is used to predict a **continuous numerical value**, such as price, salary, or temperature.

**Classification** is used to predict a **category or class**, such as Pass/Fail or Yes/No.

**Example:**

- Regression → Predict house price = ₹50 lakh
- Classification → Predict whether a student will Pass or Fail

## 3. Explain Simple Linear Regression.

Simple Linear Regression finds the relationship between **one input variable (X)** and **one output variable (Y)**.

It uses a **straight line** to make predictions.

**Formula:**  $Y = mX + c$

**Example:** Predicting marks based on hours studied.

## 4. List and explain the Assumptions of Linear Regression.

The main assumptions of Linear Regression are:

1. **Linearity** – There should be a linear relationship between X and Y.
2. **Independence** – The data observations should be independent of each other.
3. **Constant Variance** – The errors should have a similar spread.
4. **Normality of Errors** – The errors should be approximately normally distributed.
5. **No Strong Outliers** – Extreme values should not strongly affect the model.

## 5. What is Bias–Variance Trade-Off?

**Bias** is the error caused when a model is too simple and cannot learn the data properly.

**Variance** is the error caused when a model is too complex and learns the training data too much.

The **Bias–Variance Trade-Off** means finding the right balance between bias and variance.

## 6. Explain Overfitting and Underfitting with examples.

**Overfitting** happens when a model learns the training data **too much**, including noise, and performs poorly on new data.

**Underfitting** happens when a model is **too simple** and cannot learn the important patterns.

### Example:

- Overfitting → A student memorizes practice questions but cannot solve new questions.
- Underfitting → A student studies very little and cannot solve even basic questions.